مكتب

Lab Activity

Week 2: Loop Control & Game Loops

Duration: 60 minutes — Work in pairs!

Name: ____________________ **Partner:** ____________________

Lab Goals

By the end of this lab, you will be able to:

- Use `break` to exit loops early
- Use `continue` to skip parts of loops
- Build simple game menus
- Fix infinite loops

Getting Started

1. Extract the given `week2_lab.zip` file to your desktop
2. Remember: `break` = exit, `continue` = skip!

1 Exercise 1: Meet the Break Statement (7 minutes)

Open `week2_ex1.py` and type this code:

```
count = 0
while count < 10:
    count = count + 1
    if count == 5:
        break
    print(count)
print("Done!")
```

Before running: What numbers will print? ____________________

Run it. Were you right? __________

Now change it:

- Change 5 to 3. What prints now? ____________________
- Remove the `break` line. What happens? ____________________

Tip: `break` exits the loop immediately!

Checkpoint: Show your teacher the output!

2 Exercise 2: Meet Continue (8 minutes)

Open `week2_ex2.py` and type:

```
num = 0
while num < 5:
    num = num + 1
    if num == 3:
        continue
    print(num)
print("Done!")
```

Predict: Which number will NOT print? __________

Run it. Were you right? __________

Fill in this trace table:

num	num == 3?	Action
1	No	Print 1
2	No	Print 2
3	Yes	__________
4	__________	__________
5	__________	__________

Pair Programming: Switch who types every 5 minutes!

3 Exercise 3: Simple Menu (10 minutes)

Open `week2_ex3.py` and create this menu:

```
while True:
    print("\n1. Play")
    print("2. Help")
    print("3. Quit")

    choice = input("Choose: ")

    if choice == "1":
        print("Playing...")
    elif choice == "2":
        print("Press 1 to play!")
    elif choice == "3":
        print("Bye!")
        break
    else:
        print("Invalid!")
        continue
```

Test it:

- Choose 1 - does it say "Playing..."?
- Choose 3 - does it exit?
- Type "abc" - does it say "Invalid!"?

Checkpoint: Menu working? Great!

4 Exercise 4: Password Checker (8 minutes)

Complete this code in `week2_ex4.py`:

```
password = "python"
tries = 0

while tries < 3:
    tries = tries + 1
    guess = input("Password: ")

    if guess == password:
        print("Correct!")
        ______  # Exit loop
    else:
        print("Wrong!")

print("Done!")
```

Fill in the blank with the right command!

Test cases:

- Enter wrong password 3 times - should end
- Enter correct password on try 2 - should exit early

5 Exercise 5: Fix the Infinite Loops! (10 minutes)

Bug 1:

```
# This runs forever!
x = 1
while x < 5:
    print(x)
    # Missing something here!
```

Fix: Add line: ____________________

Bug 2:

```
# Never stops!
while True:
    ans = input("Type stop: ")
    if ans == "STOP":  # Bug here!
        break
```

Problem: User types "stop" but code checks for _______________

Fix: Change "STOP" to __________

Checkpoint: Both bugs fixed? Show your teacher!

6 Exercise 6: Number Game (12 minutes)

Create a simple guessing game in `week2_ex6.py`:

```
secret = 7
tries = 0

while tries < 3:
    tries = tries + 1
    guess = int(input("Guess 1-10: "))

    if guess == secret:
        print("You win!")
        break
    elif guess < secret:
        print("Higher!")
    else:
        print("Lower!")

print("Game over!")
```

Test your game:

- Guess 5 - should say "Higher!"
- Guess 9 - should say "Lower!"
- Guess 7 - should say "You win!"

Add one feature:

- Tell them how many tries they have left

Pair Programming: Switch who types every 5 minutes!

Checkpoint: Game working? Excellent!

7 Exercise 7: Skip Even Numbers (5 minutes)

Write code that prints only odd numbers from 1 to 10:

```
num = 0
while num < 10:
    num = num + 1
    if num % 2 == 0:
        _______  # Skip even numbers
    print(num)
```

Fill in the blank!

Output should be: 1, 3, 5, 7, 9

8 Lab Summary

Checkpoint: Final checkpoint - make sure you have:

- ☐ Used `break` to exit loops
- ☐ Used `continue` to skip
- ☐ Built a working menu
- ☐ Fixed infinite loops
- ☐ Created the number game
- ☐ Shown all checkpoints to teacher

Reflection (3 minutes)

Rate your understanding:

Concept	Need Help	Getting It	Got It!
break	○	○	○
continue	○	○	○
Fixing infinite loops	○	○	○

One thing I learned: ____________________________________
One thing I'm confused about: ____________________________________

Great work with loop control!